多元线性回归

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线性回归

让我们回到两个关键变量：因变量“工资”和自变量“性别”。

```
lm(HOURPAY ~ SEX, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-16.033  -7.639  -3.384   4.256  79.396

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  22.4834     0.1959  114.76   <2e-16 ***
SEXFemale    -3.0591     0.2676  -11.43   <2e-16 ***
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1

Residual standard error: 11.44 on 7345 degrees of freedom
Multiple R-squared:  0.01748,   Adjusted R-squared:  0.01734
F-statistic: 130.6 on 1 and 7345 DF,  p-value: < 2.2e-16
```

与之前一样，我们可以看到男性与女性之间的工资差异。女性的平均时薪比男性低约 £3.06。这一差异在统计上具有显著性。性别能够解释约 1.8% 的工资变异。

我们之前提出，工资差异可能还有其他解释因素。样本中的男性和女性可能在其他方面存在差异，而这些差异有助于解释为什么他们的工资不同。

我们可以通过将这些因素加入模型来控制它们的影响。例如，假设男性和女性的教育水平有所不同，这可能有助于解释工资差异。现在让我们在模型中控制教育水平。

```
lm(HOURPAY ~ SEX + HIQUL22D, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-20.566  -6.337  -2.241   3.373  77.020

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)
(Intercept)                     27.1162     0.2285  118.65   <2e-16 ***
SEXFemale                       -3.6690     0.2491  -14.73   <2e-16 ***
HIQUL22DHigher education        -5.5659     0.4812  -11.57   <2e-16 ***
HIQUL22DA level or equivalent   -7.9860     0.3203  -24.93   <2e-16 ***
HIQUL22DGCSE A*-C or equivalent -9.0507     0.3666  -24.69   <2e-16 ***
HIQUL22DOther qualitfication    -9.7761     0.6109  -16.00   <2e-16 ***
HIQUL22DNo Qualification       -11.7713     0.7544  -15.60   <2e-16 ***
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1

Residual standard error: 10.62 on 7340 degrees of freedom
Multiple R-squared:  0.1543,    Adjusted R-squared:  0.1536
F-statistic: 223.2 on 6 and 7340 DF,  p-value: < 2.2e-16
```

我们仍然可以采用与之前大致相同的方式解释模型。这里有两个分类变量。对于每个分类变量，都有一个类别被省略，作为其他类别进行比较的基准。我们称之为“参照类别”(reference category)。在本例中，性别变量的参照类别是男性，教育水平变量的参照类别是拥有学位者。

- 没有任何学历资格的人，其平均时薪比拥有学位的人低 £11.77。
- 即使教育水平相同，女性的平均时薪仍然比男性低 £3.67。

最后这一点非常重要。在第一个模型中，性别工资差距为 £3.06。然而，当我们考虑教育水平的差异后，这一差距反而扩大了。看来，样本中女性的平均教育水平高于男性。按此推测，我们原本会预期女性收入更高，但实际情况并非如此。

现在，模型能够解释工资总变异的大约 15%。看来，教育水平是决定工资的重要因素。

让我们再加入几个变量。我们可以考察一个人是否签订零工时合同（zero-hours contract，变量 FLEX22W6），以及其工作方式是兼职还是全职（FTPT）。

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-20.379  -6.206  -2.200   3.361  77.630

Coefficients:
                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                      22.0827     0.8968  24.624  < 2e-16 ***
SEXFemale                        -3.0794     0.2577 -11.949  < 2e-16 ***
HIQUL22DHigher education         -5.4010     0.4780 -11.300  < 2e-16 ***
HIQUL22DA level or equivalent    -7.7981     0.3185 -24.485  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent  -8.7889     0.3652 -24.064  < 2e-16 ***
HIQUL22DOther qualitfication     -9.5050     0.6071 -15.658  < 2e-16 ***
HIQUL22DNo Qualification        -11.3429     0.7506 -15.112  < 2e-16 ***
FLEX22W6No                       5.2259     0.8751   5.972 2.46e-09 ***
FTPTPart-time                    -2.3996     0.3164  -7.584 3.77e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.54 on 7338 degrees of freedom
Multiple R-squared:  0.1668,    Adjusted R-squared:  0.1659
F-statistic: 183.6 on 8 and 7338 DF,  p-value: < 2.2e-16
```

没有签订零工时合同的员工，平均时薪比签订此类合同的员工高 £5.23。兼职员工的平均时薪比全职员工低 £2.40。如果再次观察性别工资差距，会发现差距缩小了，现在为 £3.08。这表明，女性似乎更有可能从事兼职工作，而兼职工作的时薪又低于全职工作。

让我们尝试控制员工在现任雇主处的工作年限（EMPLEN）。

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN,
    data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-22.293  -6.130  -2.107   3.118  76.537

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)
(Intercept)                     21.0349     1.1507  18.281  < 2e-16 ***
SEXFemale                       -3.0489     0.2545 -11.978  < 2e-16 ***
HIQUL22DHigher education        -5.7451     0.4728 -12.151  < 2e-16 ***
HIQUL22DA level or equivalent   -7.8133     0.3146 -24.833  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent -9.0634     0.3613 -25.086  < 2e-16 ***
HIQUL22DOther qualitfication    -9.4442     0.5993 -15.758  < 2e-16 ***
HIQUL22DNo Qualification       -11.7892     0.7424 -15.880  < 2e-16 ***
FLEX22W6No                      3.8811     0.8717   4.452 8.63e-06 ***
FTPTPart-time                  -2.3085     0.3125  -7.387 1.67e-13 ***
EMPLENThree months, less than 6 -0.2034     0.9795  -0.208 0.835498
EMPLEN6 months, less than 12     0.4157     0.8990   0.462 0.643783
EMPLEN1 year, less than 2        0.1716     0.8205   0.209 0.834335
EMPLEN2 years, less than 5       1.4995     0.7837   1.913 0.055745 .
EMPLEN5 years, less than 10      2.8545     0.7971   3.581 0.000344 ***
EMPLEN10 years, less than 20     3.9389     0.7935   4.964 7.06e-07 ***
EMPLEN20 years or more           5.1767     0.8105   6.387 1.80e-10 ***
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1

Residual standard error: 10.4 on 7331 degrees of freedom
Multiple R-squared:  0.189, Adjusted R-squared:  0.1874
F-statistic: 113.9 on 15 and 7331 DF,  p-value: < 2.2e-16
```

在现任雇主处的某些工作年限类别，其回归系数在统计上并不显著。看来，工作年限不足 5 年的不同类别之间差别不大；但工作年限更长似乎会带来一定的工资收益。我们估计的性别工资差距基本没有变化。

现在控制工作的性质是长期职位还是临时职位（JOBTYP）。

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-22.291  -6.126  -2.103   3.113  76.546

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)
(Intercept)                     20.9736     1.1793  17.784  < 2e-16 ***
SEXFemale                       -3.0488     0.2546 -11.977  < 2e-16 ***
HIQUL22DHigher education        -5.7434     0.4729 -12.146  < 2e-16 ***
HIQUL22DA level or equivalent   -7.8108     0.3148 -24.810  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent -9.0590     0.3618 -25.039  < 2e-16 ***
HIQUL22DOther qualitfication    -9.4414     0.5995 -15.749  < 2e-16 ***
HIQUL22DNo Qualification       -11.7870     0.7425 -15.874  < 2e-16 ***
FLEX22W6No                      3.9228     0.8893   4.411 1.04e-05 ***
FTPTPart-time                  -2.3147     0.3136  -7.381 1.75e-13 ***
EMPLENThree months, less than 6 -0.2085     0.9798  -0.213 0.831462
EMPLEN6 months, less than 12     0.4261     0.9002   0.473 0.635947
EMPLEN1 year, less than 2        0.1803     0.8214   0.220 0.826233
EMPLEN2 years, less than 5       1.5153     0.7866   1.926 0.054088 .
EMPLEN5 years, less than 10      2.8723     0.8007   3.587 0.000336 ***
EMPLEN10 years, less than 20     3.9573     0.7973   4.963 7.09e-07 ***
EMPLEN20 years or more           5.1949     0.8142   6.380 1.87e-10 ***
JOBTYPNot permanent in some way  0.1705     0.7174   0.238 0.812127
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1

Residual standard error: 10.4 on 7330 degrees of freedom
Multiple R-squared:  0.189, Adjusted R-squared:  0.1873
F-statistic: 106.8 on 16 and 7330 DF,  p-value: < 2.2e-16
```

看来，这个因素在统计上并不显著。

一些证据表明，企业规模（MPNR02）可能会产生影响。让我们尝试将它加入模型。

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02, data = lfs) %>% summa
ry
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNRO2, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-24.110  -6.028  -2.068   3.031  77.222

Coefficients:
                                    Estimate Std. Error t value Pr(>|t|)
(Intercept)                         20.84866    1.20417  17.314  < 2e-16 ***
SEXFemale                           -3.02767    0.25204 -12.012  < 2e-16 ***
HIQUL22DHigher education            -5.41810    0.46892 -11.554  < 2e-16 ***
HIQUL22DA level or equivalent       -7.37501    0.31388 -23.496  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent     -8.49001    0.36100 -23.518  < 2e-16 ***
HIQUL22DOther qualitfication        -8.83659    0.59560 -14.836  < 2e-16 ***
HIQUL22DNo Qualification           -11.24586    0.73622 -15.275  < 2e-16 ***
FLEX22W6No                          3.32410    0.88173   3.770 0.000165 ***
FTPTPart-time                      -1.86697    0.31327  -5.960 2.65e-09 ***
EMPLENThree months, less than 6    -0.34450    0.97040  -0.355 0.722596
EMPLEN6 months, less than 12        0.33142    0.89180   0.372 0.710179
EMPLEN1 year, less than 2          -0.02982    0.81356  -0.037 0.970762
EMPLEN2 years, less than 5          1.25437    0.77936   1.609 0.107551
EMPLEN5 years, less than 10         2.45569    0.79377   3.094 0.001984 **
EMPLEN10 years, less than 20        3.43153    0.79075   4.340 1.45e-05 ***
EMPLEN20 years or more              4.54149    0.80804   5.620 1.98e-08 ***
JOBTYPNot permanent in some way    -0.06978    0.71049  -0.098 0.921763
MPNRO211-19                        -1.11462    0.52949  -2.105 0.035319 *
MPNRO220-24                        -0.71904    0.68668  -1.047 0.295076
MPNRO2Don't know but under 25      -2.23238    0.89162  -2.504 0.012310 *
MPNRO225-49                        -0.72845    0.46038  -1.582 0.113631
MPNRO250-249                        0.09719    0.39421   0.247 0.805267
MPNRO2250-499                       1.86456    0.56304   3.312 0.000932 ***
MPNRO2Don't know but between 50 and 499   0.78142    0.65889   1.186 0.235672
MPNRO2500 or more                   3.19590    0.39315   8.129 5.05e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.29 on 7322 degrees of freedom
Multiple R-squared:  0.2071,    Adjusted R-squared:  0.2045
F-statistic:  79.7 on 24 and 7322 DF,  p-value: < 2.2e-16
```

我们已经加入了许多分类变量。请注意，其中不少类别的系数在统计上并不显著。规模较大的企业似乎支付更高的工资，不过模型额外解释的工资变异并没有增加太多。

接下来尝试加入婚姻状况（MARSTA）。

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNRO2 + MARSTA, data = lfs)
%>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02 + MARSTA, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-24.798  -6.026  -1.954   3.214  76.056

Coefficients:
                                                 Estimate Std. Error
(Intercept)                                    19.919009   1.195219
SEXFemale                                      -3.006896   0.250727
HIQUL22DHigher education                       -5.492173   0.464690
HIQUL22DA level or equivalent                  -7.133012   0.311736
HIQUL22DGCSE A*-C or equivalent                -8.287454   0.358561
HIQUL22DOther qualitfication                   -9.007106   0.590139
HIQUL22DNo Qualification                      -11.161196   0.729359
FLEX22W6No                                     2.822237   0.873588
FTPTPart-time                                  -2.147829   0.311183
EMPLENThree months, less than 6                -0.385802   0.960456
EMPLEN6 months, less than 12                    0.100754   0.882826
EMPLEN1 year, less than 2                      -0.287125   0.805630
EMPLEN2 years, less than 5                      0.870865   0.771994
EMPLEN5 years, less than 10                     1.835654   0.787130
EMPLEN10 years, less than 20                    2.598980   0.785419
EMPLEN20 years or more                          3.467156   0.804347
JOBTYPNot permanent in some way                -0.085073   0.703213
MPNR0211-19                                    -1.031462   0.524134
MPNR0220-24                                    -0.625892   0.679732
MPNR02Don't know but under 25                  -2.135717   0.882739
MPNR0225-49                                    -0.670691   0.455719
MPNR0250-249                                    0.004716   0.390188
MPNR02250-499                                   1.800051   0.557272
MPNR02Don't know but between 50 and 499         0.753469   0.652287
MPNR02500 or more                               3.216213   0.389263
MARSTAMarried, living with spouse               3.173124   0.263834
MARSTAMarried separated from spouse             1.766559   0.816595
MARSTADivorced                                  3.415284   0.484361
MARSTAWidowed                                   3.524637   1.207088
MARSTACurrently or previously in civil partnership   0.154012   1.179948
                                                t value Pr(>|t|)
(Intercept)                                      16.666  < 2e-16 ***
SEXFemale                                       -11.993  < 2e-16 ***
HIQUL22DHigher education                        -11.819  < 2e-16 ***
HIQUL22DA level or equivalent                   -22.882  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent                 -23.113  < 2e-16 ***
HIQUL22DOther qualitfication                    -15.263  < 2e-16 ***
HIQUL22DNo Qualification                        -15.303  < 2e-16 ***
FLEX22W6No                                        3.231 0.001241 **
FTPTPart-time                                    -6.902 5.55e-12 ***
EMPLENThree months, less than 6                  -0.402 0.687927
EMPLEN6 months, less than 12                      0.114 0.909141
EMPLEN1 year, less than 2                        -0.356 0.721553
EMPLEN2 years, less than 5                        1.128 0.259326
```

```
EMPLEN5 years, less than 10                          2.332 0.019723 *
EMPLEN10 years, less than 20                         3.309 0.000941 ***
EMPLEN20 years or more                               4.311 1.65e-05 ***
JOBTYPNot permanent in some way                     -0.121 0.903712
MPNR0211-19                                         -1.968 0.049113 *
MPNR0220-24                                         -0.921 0.357189
MPNR02Don't know but under 25                       -2.419 0.015569 *
MPNR0225-49                                         -1.472 0.141139
MPNR0250-249                                         0.012 0.990357
MPNR02250-499                                        3.230 0.001243 **
MPNR02Don't know but between 50 and 499              1.155 0.248080
MPNR02500 or more                                    8.262  < 2e-16 ***
MARSTAMarried, living with spouse                   12.027  < 2e-16 ***
MARSTAMarried separated from spouse                  2.163 0.030549 *
MARSTADivorced                                       7.051 1.94e-12 ***
MARSTAWidowed                                        2.920 0.003512 **
MARSTACurrently or previously in civil partnership   0.131 0.896155
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.19 on 7317 degrees of freedom
Multiple R-squared:  0.2241,    Adjusted R-squared:  0.221
F-statistic: 72.85 on 29 and 7317 DF,  p-value: < 2.2e-16
```

婚姻状况的某些类别在统计上具有显著性。请注意，我们估计的性别工资差距有所缩小。接下来，让我们尝试一些理论上应该非常重要的因素，首先是所属行业（INDE07M）。

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02 + MARSTA + INDE07M, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-27.805  -5.904  -1.654   3.339  77.322

Coefficients:
                                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                                       18.07111    2.07800   8.696  < 2e-16 ***
SEXFemale                                         -2.12838    0.25647  -8.299  < 2e-16 ***
HIQUL22DHigher education                          -5.26106    0.45578 -11.543  < 2e-16 ***
HIQUL22DA level or equivalent                     -7.11075    0.30890 -23.020  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent                   -8.19836    0.35614 -23.020  < 2e-16 ***
HIQUL22DOther qualitfication                      -8.99971    0.58280 -15.442  < 2e-16 ***
HIQUL22DNo Qualification                         -11.44694    0.71833 -15.935  < 2e-16 ***
FLEX22W6No                                        2.13716    0.85560   2.498 0.012517 *
FTPTPart-time                                     -1.30824    0.30956  -4.226 2.41e-05 ***
EMPLENThree months, less than 6                   -0.56519    0.93856  -0.602 0.547066
EMPLEN6 months, less than 12                       0.15466    0.86267   0.179 0.857720
EMPLEN1 year, less than 2                         -0.34300    0.78733  -0.436 0.663105
EMPLEN2 years, less than 5                         0.83877    0.75470   1.111 0.266434
EMPLEN5 years, less than 10                        1.79826    0.76963   2.337 0.019490 *
EMPLEN10 years, less than 20                       2.60800    0.76807   3.396 0.000689 ***
EMPLEN20 years or more                             3.69426    0.78725   4.693 2.75e-06 ***
JOBTYPNot permanent in some way                    0.51033    0.68856   0.741 0.458621
MPNR0211-19                                       -0.50335    0.51319  -0.981 0.326710
MPNR0220-24                                       -0.01248    0.66650  -0.019 0.985057
MPNR02Don't know but under 25                     -1.40105    0.86441  -1.621 0.105102
MPNR0225-49                                        0.03596    0.44944   0.080 0.936228
MPNR0250-249                                       0.49980    0.38521   1.297 0.194509
MPNR02250-499                                      1.94956    0.54714   3.563 0.000369 ***
MPNR02Don't know but between 50 and 499            1.00554    0.64016   1.571 0.116279
MPNR02500 or more                                  3.73024    0.38657   9.650  < 2e-16 ***
MARSTAMarried, living with spouse                  3.04695    0.25907  11.761  < 2e-16 ***
MARSTAMarried separated from spouse                1.84784    0.79818   2.315 0.020637 *
MARSTADivorced                                     3.52986    0.47429   7.442 1.10e-13 ***
MARSTAWidowed                                      3.68233    1.18007   3.120 0.001813 **
MARSTACurrently or previously in civil partnership 0.16740    1.15310   0.145 0.884575
INDE07MEnergy and water                            4.42746    1.91293   2.314 0.020668 *
INDE07MManufacturing                               2.28935    1.78234   1.284 0.199019
INDE07MConstruction                                4.48751    1.82011   2.466 0.013705 *
INDE07MDistribution, hotels and restaurants       -0.81363    1.76477  -0.461 0.644784
INDE07MTransport and communication                 4.96552    1.77793   2.793 0.005238 **
INDE07MBanking and finance                         4.19420    1.75992   2.383 0.017189 *
INDE07MPublic admin, education and health         -0.39858    1.75291  -0.227 0.820134
INDE07MOther services                             -0.11917    1.82864  -0.065 0.948042
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.952 on 7309 degrees of freedom
Multiple R-squared:  0.2602,    Adjusted R-squared:  0.2565
F-statistic: 69.48 on 37 and 7309 DF,  p-value: < 2.2e-16
```

看来，不同行业之间确实存在工资差异。一些男性占主导地位的行业似乎支付更高的工资。请注意，控制行业因素后，我们对性别工资差距的估计进一步缩小到 £2.13，明显低于最初的估计。接下来，尝试控制职业等级或资历层级（NSECMJ20）。

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNRO2 + MARSTA + INDEO7M + N
SECMJ20, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-29.785  -5.085  -1.209   2.884  79.650

Coefficients:
                                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                                      26.19639    1.93934  13.508  < 2e-16 ***
SEXFemale                                        -1.81276    0.24012  -7.550 4.90e-14 ***
HIQUL22DHigher education                         -3.17471    0.42574  -7.457 9.88e-14 ***
HIQUL22DA level or equivalent                    -4.11659    0.29875 -13.779  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent                  -4.40433    0.34803 -12.655  < 2e-16 ***
HIQUL22DOther qualitfication                     -4.10139    0.56512  -7.258 4.35e-13 ***
HIQUL22DNo Qualification                         -5.92687    0.69490  -8.529  < 2e-16 ***
FLEX22W6No                                        0.39773    0.79824   0.498 0.618316
FTPTPart-time                                     0.03967    0.29256   0.136 0.892142
EMPLENThree months, less than 6                  -0.44517    0.86747  -0.513 0.607840
EMPLEN6 months, less than 12                     -0.30842    0.79761  -0.387 0.699004
EMPLEN1 year, less than 2                        -0.48591    0.72751  -0.668 0.504219
EMPLEN2 years, less than 5                        0.35393    0.69807   0.507 0.612162
EMPLEN5 years, less than 10                       0.71789    0.71281   1.007 0.313906
EMPLEN10 years, less than 20                      1.51428    0.71140   2.129 0.033322 *
EMPLEN20 years or more                            2.41304    0.72966   3.307 0.000947 ***
JOBTYPNot permanent in some way                   0.21536    0.63742   0.338 0.735473
MPNR0211-19                                      -0.14893    0.47435  -0.314 0.753551
MPNR0220-24                                       0.11269    0.61576   0.183 0.854796
MPNR02Don't know but under 25                    -1.16732    0.79908  -1.461 0.144106
MPNR0225-49                                      -0.08240    0.41601  -0.198 0.843000
MPNR0250-249                                      0.35521    0.35649   0.996 0.319086
MPNR02250-499                                     1.37499    0.50709   2.712 0.006713 **
MPNR02Don't know but between 50 and 499           0.67591    0.59225   1.141 0.253801
MPNR02500 or more                                 2.30709    0.36185   6.376 1.93e-10 ***
MARSTAMarried, living with spouse                 2.34713    0.24139   9.724  < 2e-16 ***
MARSTAMarried separated from spouse               1.98767    0.73805   2.693 0.007094 **
MARSTADivorced                                    3.18823    0.43928   7.258 4.34e-13 ***
MARSTAWidowed                                     3.32084    1.09141   3.043 0.002353 **
MARSTACurrently or previously in civil partnership  0.07321  1.06572   0.069 0.945237
INDE07MEnergy and water                           3.76222    1.76919   2.127 0.033494 *
INDE07MManufacturing                              2.05983    1.64759   1.250 0.211262
INDE07MConstruction                               3.77356    1.68292   2.242 0.024974 *
INDE07MDistribution, hotels and restaurants      -0.47190    1.63117  -0.289 0.772358
INDE07MTransport and communication                3.96325    1.64482   2.410 0.015998 *
INDE07MBanking and finance                        2.85373    1.62740   1.754 0.079550 .
INDE07MPublic admin, education and health        -0.32214    1.61998  -0.199 0.842384
INDE07MOther services                            -0.36766    1.69012  -0.218 0.827796
NSECMJ20Lower managerial and professional        -6.10448    0.29727 -20.535  < 2e-16 ***
NSECMJ20Intermediate occupations                -10.73648    0.37507 -28.625  < 2e-16 ***
NSECMJ20Lower supervisory and technical         -10.11014    0.52727 -19.175  < 2e-16 ***
NSECMJ20Semi-routine occupations                -11.83911    0.45612 -25.956  < 2e-16 ***
NSECMJ20Routine occupations                     -12.62290    0.47731 -26.446  < 2e-16 ***
NSECMJ20Never worked                            -10.00244    0.81956 -12.205  < 2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.194 on 7303 degrees of freedom
Multiple R-squared:  0.3691,    Adjusted R-squared:  0.3654
F-statistic: 99.38 on 43 and 7303 DF,  p-value: < 2.2e-16
```

不同职业等级之间存在非常明显的差异。例如，从事常规性职业（routine occupations）的人员，其平均时薪比高级管理职位人员低约 £12.62。加入这一因素后，估计的性别工资差距进一步缩小到 £1.81。

让我们尝试在回归模型中加入年龄变量 AGE。

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNRO2 + MARSTA + INDE07M + N
SECMJ20 + AGE, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 + AGE, data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-30.360  -5.126  -1.250   2.981  80.029

Coefficients:
                                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                                      2.274e+01  1.963e+00  11.584  < 2e-16 ***
SEXFemale                                       -1.753e+00  2.388e-01  -7.339 2.39e-13 ***
HIQUL22DHigher education                        -3.374e+00  4.238e-01  -7.961 1.96e-15 ***
HIQUL22DA level or equivalent                   -4.115e+00  2.970e-01 -13.856  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent                 -4.799e+00  3.486e-01 -13.766  < 2e-16 ***
HIQUL22DOther qualitfication                    -4.428e+00  5.629e-01  -7.866 4.20e-15 ***
HIQUL22DNo Qualification                        -6.519e+00  6.938e-01  -9.396  < 2e-16 ***
FLEX22W6No                                       2.230e-01  7.938e-01   0.281 0.778816
FTPTPart-time                                   -9.227e-02  2.912e-01  -0.317 0.751351
EMPLENThree months, less than 6                 -4.469e-01  8.624e-01  -0.518 0.604330
EMPLEN6 months, less than 12                    -3.018e-01  7.930e-01  -0.381 0.703485
EMPLEN1 year, less than 2                       -4.897e-01  7.233e-01  -0.677 0.498380
EMPLEN2 years, less than 5                       1.750e-01  6.943e-01   0.252 0.800970
EMPLEN5 years, less than 10                      3.528e-01  7.098e-01   0.497 0.619169
EMPLEN10 years, less than 20                     8.021e-01  7.114e-01   1.127 0.259590
EMPLEN20 years or more                           1.112e+00  7.388e-01   1.505 0.132267
JOBTYPNot permanent in some way                  2.220e-01  6.337e-01   0.350 0.726105
MPNR0211-19                                     -3.772e-05  4.719e-01   0.000 0.999936
MPNR0220-24                                      1.658e-01  6.122e-01   0.271 0.786569
MPNR02Don't know but under 25                   -9.276e-01  7.949e-01  -1.167 0.243226
MPNR0225-49                                      2.462e-02  4.137e-01   0.059 0.952558
MPNR0250-249                                     5.388e-01  3.550e-01   1.518 0.129071
MPNR02250-499                                    1.496e+00  5.043e-01   2.967 0.003016 **
MPNR02Don't know but between 50 and 499          7.647e-01  5.889e-01   1.299 0.194123
MPNR02500 or more                                2.556e+00  3.607e-01   7.086 1.51e-12 ***
MARSTAMarried, living with spouse                1.446e+00  2.588e-01   5.589 2.36e-08 ***
MARSTAMarried separated from spouse              8.427e-01  7.440e-01   1.133 0.257420
MARSTADivorced                                   1.729e+00  4.640e-01   3.727 0.000195 ***
MARSTAWidowed                                    1.635e+00  1.100e+00   1.487 0.137187
MARSTACurrently or previously in civil partnership -6.873e-01  1.063e+00  -0.647 0.517819
INDE07MEnergy and water                          3.754e+00  1.759e+00   2.134 0.032857 *
INDE07MManufacturing                             2.071e+00  1.638e+00   1.264 0.206146
INDE07MConstruction                              3.781e+00  1.673e+00   2.260 0.023865 *
INDE07MDistribution, hotels and restaurants     -3.048e-01  1.622e+00  -0.188 0.850906
INDE07MTransport and communication               4.013e+00  1.635e+00   2.454 0.014148 *
INDE07MBanking and finance                       2.905e+00  1.618e+00   1.795 0.072643 .
INDE07MPublic admin, education and health       -4.107e-01  1.611e+00  -0.255 0.798724
INDE07MOther services                           -3.300e-01  1.680e+00  -0.196 0.844317
NSECMJ20Lower managerial and professional       -6.054e+00  2.956e-01 -20.479  < 2e-16 ***
NSECMJ20Intermediate occupations                -1.069e+01  3.729e-01 -28.676  < 2e-16 ***
NSECMJ20Lower supervisory and technical         -1.001e+01  5.243e-01 -19.101  < 2e-16 ***
NSECMJ20Semi-routine occupations                -1.180e+01  4.535e-01 -26.015  < 2e-16 ***
NSECMJ20Routine occupations                     -1.275e+01  4.747e-01 -26.851  < 2e-16 ***
NSECMJ20Never worked                            -8.792e+00  8.251e-01 -10.655  < 2e-16 ***
```

```
AGE                                          1.071e-01  1.151e-02   9.306  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.14 on 7302 degrees of freedom
Multiple R-squared:  0.3765,    Adjusted R-squared:  0.3728
F-statistic: 100.2 on 44 and 7302 DF,  p-value: < 2.2e-16
```

年龄每增加 1 岁，与平均时薪提高 £0.11 相关。在控制年龄后，我们估计的性别工资差距进一步缩小，目前为 £1.75。最后，让我们尝试控制地区差异（GOR9D）。

```
lm(HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN + JOBTYP + MPNR02 + MARSTA + INDE07M + N
SECMJ20 + AGE + GOR9D, data = lfs) %>% summary
```

```
Call:
lm(formula = HOURPAY ~ SEX + HIQUL22D + FLEX22W6 + FTPT + EMPLEN +
    JOBTYP + MPNR02 + MARSTA + INDE07M + NSECMJ20 + AGE + GOR9D,
    data = lfs)

Residuals:
    Min      1Q  Median      3Q     Max
-31.140  -5.101  -1.189   3.046  79.547

Coefficients:
                                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                                     25.78853    1.98215  13.010  < 2e-16 ***
SEXFemale                                       -1.78189    0.23677  -7.526 5.87e-14 ***
HIQUL22DHigher education                        -3.23076    0.42198  -7.656 2.16e-14 ***
HIQUL22DA level or equivalent                   -3.85940    0.29611 -13.034  < 2e-16 ***
HIQUL22DGCSE A*-C or equivalent                 -4.53640    0.34719 -13.066  < 2e-16 ***
HIQUL22DOther qualitfication                    -4.44594    0.55803  -7.967 1.87e-15 ***
HIQUL22DNo Qualification                        -6.14744    0.68828  -8.932  < 2e-16 ***
FLEX22W6No                                       0.40302    0.78712   0.512 0.608648
FTPTPart-time                                   -0.07943    0.28902  -0.275 0.783459
EMPLENThree months, less than 6                 -0.39861    0.85519  -0.466 0.641152
EMPLEN6 months, less than 12                    -0.18981    0.78648  -0.241 0.809302
EMPLEN1 year, less than 2                       -0.30792    0.71735  -0.429 0.667754
EMPLEN2 years, less than 5                       0.34445    0.68859   0.500 0.616933
EMPLEN5 years, less than 10                      0.52264    0.70371   0.743 0.457692
EMPLEN10 years, less than 20                     1.06104    0.70594   1.503 0.132879
EMPLEN20 years or more                           1.57930    0.73389   2.152 0.031434 *
JOBTYPNot permanent in some way                  0.15802    0.62841   0.251 0.801461
MPNR0211-19                                      0.01830    0.46778   0.039 0.968800
MPNR0220-24                                      0.17017    0.60750   0.280 0.779394
MPNR02Don't know but under 25                   -1.01501    0.78914  -1.286 0.198407
MPNR0225-49                                      0.05431    0.41028   0.132 0.894685
MPNR0250-249                                     0.51390    0.35217   1.459 0.144539
MPNR02250-499                                    1.50815    0.50016   3.015 0.002576 **
MPNR02Don't know but between 50 and 499          0.65691    0.58493   1.123 0.261452
MPNR02500 or more                                2.40393    0.35823   6.711 2.08e-11 ***
MARSTAMarried, living with spouse                1.44619    0.25735   5.620 1.98e-08 ***
MARSTAMarried separated from spouse              0.79898    0.73767   1.083 0.278795
MARSTADivorced                                   1.68799    0.46047   3.666 0.000248 ***
MARSTAWidowed                                    1.62508    1.09137   1.489 0.136521
MARSTACurrently or previously in civil partnership -0.38073  1.05538  -0.361 0.718297
INDE07MEnergy and water                          3.70456    1.74382   2.124 0.033671 *
INDE07MManufacturing                             2.19235    1.62412   1.350 0.177099
INDE07MConstruction                              3.84059    1.65911   2.315 0.020649 *
INDE07MDistribution, hotels and restaurants     -0.39967    1.60802  -0.249 0.803718
INDE07MTransport and communication               3.64347    1.62184   2.246 0.024702 *
INDE07MBanking and finance                       2.61233    1.60432   1.628 0.103503
INDE07MPublic admin, education and health       -0.37672    1.59701  -0.236 0.813522
INDE07MOther services                           -0.61520    1.66640  -0.369 0.712006
NSECMJ20Lower managerial and professional       -5.89892    0.29352 -20.097  < 2e-16 ***
NSECMJ20Intermediate occupations               -10.52187    0.37006 -28.433  < 2e-16 ***
NSECMJ20Lower supervisory and technical         -9.81153    0.51984 -18.874  < 2e-16 ***
NSECMJ20Semi-routine occupations               -11.54158    0.45036 -25.627  < 2e-16 ***
NSECMJ20Routine occupations                    -12.49647    0.47149 -26.504  < 2e-16 ***
```

```
NSECMJ2ONever worked                    -8.66825    0.81832 -10.593  < 2e-16 ***
AGE                                      0.10231    0.01142   8.956  < 2e-16 ***
GOR9DYorkshire and The Humber           -4.31942    0.50909  -8.485  < 2e-16 ***
GOR9DSouth West                         -3.69832    0.50952  -7.258 4.32e-13 ***
GOR9DWest Midlands                      -3.25102    0.54650  -5.949 2.83e-09 ***
GOR9DSouth East                         -2.42283    0.47010  -5.154 2.62e-07 ***
GOR9DEast of England                    -2.23360    0.53480  -4.176 2.99e-05 ***
GOR9DEast Midlands                      -4.01808    0.52771  -7.614 2.99e-14 ***
GOR9DNorth East                         -5.24307    0.65972  -7.947 2.19e-15 ***
GOR9DNorth West                         -4.00454    0.52126  -7.682 1.77e-14 ***
GOR9DWales                              -4.97962    0.66622  -7.474 8.65e-14 ***
GOR9DNorthern Ireland                   -5.08488    0.57151  -8.897  < 2e-16 ***
GOR9DScotland                           -3.28323    0.53089  -6.184 6.57e-10 ***
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1

Residual standard error: 9.055 on 7291 degrees of freedom
Multiple R-squared:  0.389, Adjusted R-squared:  0.3844
F-statistic: 84.41 on 55 and 7291 DF,  p-value: < 2.2e-16
```

看来确实存在地区之间的工资差异。似乎所有地区的工资都低于伦敦。请注意，这里讨论的是名义工资（nominal wages），而不是扣除物价差异后的实际工资（real wages），这可能解释其中一部分差异。地区变量再次改变了我们对性别工资差距的估计，目前为 £1.78。现在，模型能够解释约 39% 的工资变异。虽然大部分变异仍然没有得到解释，但考虑到决定一个人工资的因素非常多，我们的模型其实仍然相当简单。